Joseph A. Kleinkopf

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Education

2027	**Ph.D., Plant Biology**, Cornell University, Ithaca, NY *Committee: Drs. Hannah Marx (PI), Daniel Anstett, Matt Pennell, Brent Sinclair*
2019	**M.Ed., Secondary Science Education**, University of Massachusetts, Boston, MA
2018	**M.S., Plant Biology**, Washington State University, Pullman, WA *Committee:* Drs. Eric Roalson (PI), Larry Hufford, Joanna Kelley, Warren Wagner
2015	**B.A., MCDB and Japanese**, University of Colorado, Boulder, CO

Research Projects

2023 -	**Biome evolution in *Erigeron* (Asteraceae; Astereae)**, Marx Lab, Cornell University *Reconstructing phylogenetic relationships in Erigeron to test hypotheses of alpine plant origins.*
2021-2026	**Drivers of alpine plant community turnover**, Marx Lab, Cornell University *Testing hypotheses of alpine plant community turnover across western North America.*
2022-2025	**Silica-dried tissue collection infrastructure**, UNM Herbarium, University of New Mexico *Developing a standardized protocol for storing and digitizing silica-dried plant tissues.*
2021-2023	**Adaptability of Piñon Pine,** Marx Lab, University of New Mexico *Transcriptome assembly and DNA probe design. Project led by Dr. Diana Macias.*
2016-2018	**Systematics of Hawaiian *Cyrtandra*,** Roalson Lab, Washington State University *Phylogenetic analysis and tests for hybridization in Hawaiian* Cyrtandra *using Genome-Wide Data.*
2014-2015	**Aberrant plant diversity**, Tripp Lab & COLO Herbarium, University of Colorado *Genetic differentiation in morphologically aberrant forms of Purgatory plant species.*

Peer-Reviewed Publications

7. Macias, D.S., E.F. Gyllenhaal, **J. Kleinkopf**, W.T. Pockman, H. Marx. **In review.** *A multi-tissue transcriptome strategy for genomic resource development in* Pinus edulis. Applications in Plant Sciences. https://doi.org/10.64898/2026.10.05.756810
6. **Kleinkopf, J.A.,** H. Faust, D.E. Giblin, M.R. Gostel, C.M. Guilliams, K.E. Hasenstab-Lehman, J.M. Heberling, S.J. Jacobs, B. Legler, R. Levy, A.G. Linan, L.C. Majure, K.D. Pearson, T.F. Rehman, C.J. Rothfels, D.C. Tank, W. Testo, C.M. Tribble, and H.E. Marx. **In press**. *Building infrastructure for silica-preserved leaf tissue collections in herbaria.* Systematic Botany.
5. Faust, H., **J.A. Kleinkopf**, B. Olmeda, and G.L. McCartha. 2026. *Updates to the flora of New Mexico (U.S.A.): Notable botanical findings from the next generation of southwestern botanists.* Journal of the Botanical Research Institute of Texas. 20(1). [Link]
4. Lombardi, E.M., B.D. Inouye, **J.A. Kleinkopf**, J.S. Reithel, E.H. Shapiro, J.A. Smith, and N. Underwood. 2025. *Climate change data in the biology classroom: three assignment sequences leverage long-term field experiments to teach about change in mountain ecosystems*. CourseSource. 12. [Link]
3. Ertsgaard, E.W., N.L. Gjording, J.D. Bakker, **J.A. Kleinkopf**, and D.E. Giblin. 2025. *Geology and climate drive alpine plant compositional variation among peaks in the Cascade Mountain Range*. PLoS One. e0317140. [Link]
2. **Kleinkopf, J.A.**, W.R. Roberts, W.H. Wagner, and E.H. Roalson. 2019. *Diversification of Hawaiian* Cyrtandra *(Gesneriaceae) under the influence of incomplete lineage sorting and hybridization*. Journal of Systematics and Evolution. 57(6). [Link]
1. **Kleinkopf, J.A.**, D.A. Clark, and E.A. Tripp. 2017. *Aberrant Plant Diversity in the Purgatory Watershed of Southeastern Colorado and Northeastern New Mexico*. Western North American Naturalist. 77(3). [Link]

Presentations

- **Kleinkopf, J.A.** September 2026. *The biodiversity and origins of the alpine plants of North America*. Presentation. North American Rock Garden Society, Adirondack Chapter, Ithaca, NY.

- **Kleinkopf, J.A.** August 2026. *Evolution of alpine Fleabanes (*Erigeron *L.) in North America*. Presentation. TICA Talks, The International Compositae Alliance, Virtual Seminar. [Link]
- **Kleinkopf, J.A.**, H.E. Marx. August 2026. *Biogeography and Biome Evolution in* Erigeron *L*. Presentation. Botany Conference, Tucson, AZ.
- **Kleinkopf, J.A.**, H.E. Marx. June 2026. *The evolution and origins of alpine daisies in North America*. Poster. Evolution Conference, Cleveland, OH.
- Marx, H.E., **J.A. Kleinkopf.** July 2025. *Diversity and drivers of alpine plant community structure in western North America.* Presentation. Botany Conference, Palm Springs, CA.
- **Kleinkopf, J. A.** July 2024. *The alpine flora of the southern Rocky Mountains: Past, present, and future.* Presentation. New Mexico Native Plant Society Conference, Taos, NM.
- **Kleinkopf, J. A.**, H.E. Marx. June 2024. *Drivers of alpine plant community turnover across Western North America.* Presentation. Botany Conference, Grand Rapids, MI.
- Westerman, J., **J.A. Kleinkopf**, H. E. Marx. June 2024. *Effects of pollination syndromes on alpine plant communities in the Southern Rocky Mountains.* Poster. Botany Conference, Grand Rapids, MI.
- **Kleinkopf, J.A.** February 2024. *The alpine flora of New Mexico.* Presentation. New Mexico Native Plant Society, Albuquerque, NM.
- **Kleinkopf, J.A.** November 2023. *The alpine flora: Where do we go from here?* Presentation. Museum Research Traineeship Seminar Series, Albuquerque, NM.
- **Kleinkopf, J.A.** October, 2022. *Testing hypotheses of North American alpine plant community assembly.* Presentation. MSB Nights Seminar Series, Albuquerque, NM.
- **Kleinkopf, J.A.**, H. E. Marx, and D. E. Giblin. July 2022. *Alpine Plant Community Phylogenetics of the Greater Pacific Northwest*. Poster. Botany Conference, Anchorage, AK.
- **Kleinkopf, J. A.**, E. H. Roalson, W. R. Roberts, and W. L. Wagner. March 2018. *Detecting Hybridization in Hawaiian* Cyrtandra *using Genome-Wide Data*. Poster. Washington State University Biology Graduate Student Symposium, Moscow, ID.

Teaching Experience

2026 Sum.	**Graduate Teaching Assistant**, L.H. Bailey Hortorium Herbarium, Cornell University *Taught student-interns plant collection techniques and herbarium best practices.*
2025 Spr.	**Teaching Assistant**, Department of Biology, U. of New Mexico *General Botany (BIOL-464L). Designed lab activities and assessed student learning.*
2023 Fall	**Instructor of Record**, Department of Biology, U. of New Mexico *General Botany Lecture and Lab (BIOL-360L, 4 credit course). Redesigned course materials to be student centered, culturally responsive, and better incorporate our LMS, Canvas.*
2019-2021	**Science Teacher**, Salish Middle School, North Thurston Public Schools, Olympia, WA *6th Grade Earth and Space Science, 6-8th Grade Advisory*
2018-2019	**Science Teacher Resident**, Boston Teacher Residency, Boston Plan for Excellence *10th Grade Biology, 11th/12th Grade Anatomy and Physiology*
2016-2018	**Teaching Assistant**, Washington State U. *General Biology and Lab (General Biology BIOL 102 and Lab BIOL 105; four semesters)*
2013 Sum.	**Undergraduate Teaching Assistant,** Science Discovery, U. of Colorado *K-12 Science Summer Camp Teaching Assistant*

Professional Experience

2025-2026 **Graduate Research Assistant,** Marx Lab, Cornell University
Alpine plant surveys and analyzing drivers of alpine plant community turnover.

2024 Fall **Research Assistant,** Marx Lab, University of New Mexico
DNA extractions and phylogenetic analysis of alpine plants/Erigeron herbarium specimens.

2022-2023 **Graduate Assistant,** Museum of Southwestern Biology Herbarium, U. of New Mexico
Inventory and reorganize Type specimens by APG IV order.
Reorganize the general collections from alphabetical to APG IV.
Begin development of infrastructure to store and database silica-dried tissue samples.

2021-2022 **Museum Research Training Program Fellow,** U. of New Mexico
Interdisciplinary program focusing on the practices and issues in collections museums.

2017 Sum. **Graduate Assistant,** Hudson Biological Reserve, Washington State U.
Survey of floral diversity at the Hudson Biological Reserve at Smoot Hill.

Fellowships and Grants

2026 $600 **Conference Grant**, Graduate School, Cornell University
Travel to Tucson, Arizona to present research findings at the Botany Conference.

2025 $17831 **USDA-ARS Fellowship**, Buckler Lab AI/CERCA, Cornell University
Using AI tools to explore biome evolution and evolution of cold-tolerance in Erigeron *L.*

2025 $3500 **Moore Graduate Research Award**, L.H. Bailey Hortorium, Cornell University
Field work to collect alpine plants across the Southern Rocky Mountains.

2024 $500 **NYBG Visiting Scientist Research Award**, New York Botanical Garden
Visit to NYBG to sample Erigeron *specimens housed at the Steere Herbarium.*

2024 $400 **GRAC Research Award**, BGSA, U. of New Mexico
Lab consumables for DNA extraction of Erigeron *herbarium specimens.*

2024 $1500 **Rogers Research Award**, GPSA, U. of New Mexico
Lab consumables for DNA extraction of Erigeron *herbarium specimens.*

2024 $1200 **Graduate Student Research Grant,** American Society of Plant Taxonomists
Origins of the Alpine Flora: Range and Biome Evolution in Conyzinae (Asteraceae: Astereae)

2024 $2000 **MRT Travel and Research Grants,** NSF Museum Research Traineeship, U. of New Mexico
Travel to Botany Conference 2024 and herbaria for destructive Erigeron *sampling.*

2024 $1500 **Grove Summer Research Scholarship,** Biology Department, U. of New Mexico
Materials for DNA extraction and sequencing for Erigeron phylogenetics.

2024 $500 **Student Research Grant,** GPSA, U. of New Mexico
Materials for alpine plant DNA extraction.

2023 $2500 **Lewis and Clark Fund for Exploration and Field Research,** American Philosophical Society
Field work to collect alpine plants across the Southern Rocky Mountains.

2023 $2000 **Grove Summer Research Scholarship,** Biology Department, U. of New Mexico
Field work to collect alpine plants across the Southern Rocky Mountains.

2023 $350 **Professional Development Grant,** GPSA, U. of New Mexico
Registration to attend the Evolution Conference in Albuquerque, NM.

2023 $1500 **Jack & Martha Carter Conservation Fund Grant,** The Native Plant Society of New Mexico
Field work to collect alpine plants across the Southern Rocky Mountains.

2022 $500 **Student Research Grant,** GPSA, U. of New Mexico
Travel to attend the Botany Conference in Anchorage, AK.

2022 $1800 **Grove Summer Research Scholarship,** Biology Department, U. of New Mexico
Field work to collect alpine plants across the Southern Rocky Mountains.

2021 $45750 **NSF Museum Research Training Fellowship,** U. of New Mexico
Tuition, fees, and stipend. Interdisciplinary program to learn about museum work.

2018 $2200 **Higinbotham Travel Grant,** Biology Department, Washington State U.
Field work to collect Cyrtandra *species on the Hawaiian Islands.*

Service and Outreach

Service

2026- **Reviewing Editor Board**, Applications in Plant Sciences, Botanical Society of America
2026- **Vice President**, Botanical Society of America Student Chapter, Cornell University
2026 Sum. **Symposium Co-Organizer**, Botany Conference 2026, Tucson, AZ
Advancing Alpine Plants as Models of Eco-Evolutionary Dynamics at the Boundaries of Plant Survival
2024-2025 **Herbarium Faculty-Curator Hiring Committee BGSA Representative**, U. of New Mexico
2021-2024 **GRAC Grant Proposal Reviewer**, BGSA, U. of New Mexico
2023-2024 **Graduate Research Allocation Committee (GRAC) Chair**, BGSA, U. of New Mexico
2022-2023 **Undergraduate Policy Committee Chair**, BGSA, U. of New Mexico
2020-2021 **Equity Committee Co-Lead**, Salish Middle School, North Thurston Public Schools
2019-2020 **Equity Committee**, Salish Middle School, North Thurston Public Schools

Outreach

2024- **Alpine Botany Journal Club Organizer**, Monthly virtual meetings
2026 Sum. **Alumni Open House,** L.H. Bailey Hortorium Herbarium, Cornell University
2024 Sum. **Native Plant Walk to Williams Lake**, NM Native Plant Society Conference
2022-2024 **Plant Identification Nights**, Museum of Southwestern Biology Herbarium (UNM)
2023-2024 **Jefferson Middle School Partnership**, Museum of Southwestern Biology
2023 Sum. **Night at the Museum (Evolution Conference 2023)**, NMMNHS
2023 Sum. **Open House (Evolution Conference 2023)**, Museum of Southwestern Biology (UNM)
Evolution Conference 2023, Community Outreach 2023
2022 Fall **Herbarium Tour, Biology Undergrad Society**, Museum of Southwestern Biology (UNM)
2021-2022 **Skype-a-Scientist: Scientist** (Virtual)
Mr. Wessel's 12th grade class, Ms. Kieser's 5th grade class, Ms. Sanyo's 4th grade class
2020-2021 **Skype-a-Scientist: Teacher** (Virtual)
Jackie Billotte (CU-Denver), Nick Gladstone (U. Tennessee), Maia Casna (Leiden University)
2017-2018 **Outdoor Family Fun Days**, BGSA, Washington State U.
2016-2017 **Biology Family Fun Days**, BGSA, Washington State U.

Guest Lectures

2025 Fall **Concepts and Techniques in Plant Biology**, PLSCI 6831, Cornell University
Lecture topic: *Homology, sequence alignment, and parsimony*
2023 Spr. **Botanical Diversity and Preservation**, BIOL 419/519, U. of New Mexico
Lecture topic: *Bias in natural history collections*

Mentoring

2026- **Leah Wong**. Undergraduate; data entry, data analysis; Cornell University
2025-2026 **Dylan Sheng**. Lab technician; lab work, data analysis; Cornell University
2022-2025 **Jeremiah Westerman**. Undergraduate; fieldwork, lab work, writing; U. of New Mexico
2023-2024 **Adrian Vigil**. Undergraduate; lab work; U. of New Mexico
2023 Sum. **Brianna Addison**. Undergraduate; field work; U. of New Mexico
2022 Sum. **Lauren Urenda**. Undergraduate; field work; University of New Mexico
2017 Sum. **Mara Huang**. Undergraduate; field work; Washington State U.
2017 Sum. **Tia Prudholm**. Undergraduate; field work; Washington State U.

Professional Society Membership

Botanical Society of America (BSA), American Society of Plant Taxonomists (ASPT), Society for the Study of Evolution (SSE), Society of Systematic Biologists (SSB), New Mexico Native Plant Society (NMNPS)

Journals Refereed

American Journal of Botany (1), Annals of Botany (1), Systematic Botany (1), Western North American Naturalist (1)

Honors and Awards

2023 **1st Place, The Faces of Biology Contest**, American Institute of Biological Sciences
Cover Image of Bioscience Vol 74, Issue 4 (April 2024) [Link]
2015 **EBIO Undergraduate Scientist Feature,** University of Colorado-Boulder
2014 **Phi Beta Kappa Honors Society,** University of Colorado-Boulder

Professional Development

2024 **Biodiversity Data Wrangling Workshop**, Botany Conference, Grand Rapids, MI
2024 **Workshop on Molecular Evolution**, Marine Biological Laboratory, Woods Hole, MA
2023 **Student Experience Project for Equitable Pedagogy**, University of New Mexico, Albuquerque, NM
2023 **RoLE Biodiversity Data Workshop**, Evolution Conference, Albuquerque, NM
2022 **Plant Collection Workshop**, Sevilleta Wildlife Refuge, Socorro, NM

Fieldwork and Herbarium Collections

Vouchered Herbarium Specimens:

2026 **Tara Atluri Internship**, Finger Lakes Region, NY (421 collections, BH Herbarium)
2025 **Alpine Plants**, Wheeler Peak Wilderness Area, NM (60 collections, UNM Herbarium)
2024 **Alpine Plants**, Pecos Wilderness Area, NM (97 collections, UNM Herbarium)
2023 **Alpine Plants**, Southern Rocky Mountains, NM & CO (601 collections, UNM Herbarium)
2022 **Alpine Plants**, Southern Rocky Mountains, NM & CO (551 collections, UNM Herbarium)
2022 **Sevilleta Wildlife Refuge**, Socorro Co., NM (2 collections, UNM Herbarium)
2017 **Hudson Biological Reserve**, Whitman Co., WA (42 collections, WS Herbarium)
2016 **Picket Wire (Purgatoire) Canyonlands**, Las Animas Co., CO (COLO Herbarium)
2016 **Comanche National Grasslands BioBlitz**, Las Animas Co., CO (COLO Herbarium)

Duplicate Specimens: BH, CAS, DBG, ID, NAU, NYBG, RM, WTU

Notable Collections:

Carex perglobosa (Cyperaceae): New Mexico state record. Kleinkopf 1182 at UNM.
Phacelia hastata var. alpina (Hydrophyllaceae): New Mexico state record. Kleinkopf 976 at UNM.

Professional References

Dr. Hannah Marx | hem73 [at] cornell.edu
Assistant Professor & Director of the L.H. Bailey Hortorium Herbarium, Cornell University (BH)
Dr. David Giblin | dgiblin [at] uw.edu
Herbarium Collections Manager, Burke Museum, University of Washington (WTU)
Dr. Eric DeChaine | dechaie [at] wwu.edu
Professor & Curator of the Pacific Northwest Herbarium, Western Washington University (WWB)
Dr. Eric Roalson | eric_roalson [at] wsu.edu
Professor & Director of the Ownbey Herbarium, Washington State University (WS)
Dr. Warren Wagner | wagnerw [at] si.edu
Research Botanist and Curator of Botany, Smithsonian Institute (US)